Disclosure Day
Compact proof that we most likely inhabit an OPH simulation

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Observer-Patch Holography (OPH) begins with one radical subtraction. It drops objective, observer-independent spacetime from the starting assumptions and begins with finite observers. Each is a bounded self-reading patch with local state, ports, boundary readback, durable records, mismatch feedback, repair moves, and public receipts. No patch contains the world. Reality is the stable public fixed point that survives when all local views are made mutually consistent,

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

This lean premise generates a surprising amount of physics. On the declared receipt branch, five observer-consistency axioms reach the Born–Lüders event surface, Lorentzian $3+1$ spacetime, Einstein dynamics, the Standard Model quotient and charge lattice, three colors, three generations, one Higgs doublet, the weak hierarchy, cosmic capacity closure, and the geometric normalization of Newton coupling. The same architecture also fixes sharp structural consequences: integer charge for color singlets, no gauge-mediated proton decay, two classical Maxwell modes, and two classical Einstein transverse-traceless modes. OPH functions as a candidate generative compression of essentially every foundational layer of modern physics.

Its case rests on reuse: the same observer-patch normal form yields quantum readout and the conditional Lorentz–Einstein chain; the same compact-gauge branch yields the Standard Model quotient, matter lattice, and $(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4)$; the same pixel endpoint enters the source-side electromagnetic root and weak hierarchy; and the same edge-entropy normalization fixes geometric Newton coupling. The rest of this note displays that chain, result by result. Here "most likely" refers to the joint overconstraint and coincidence budget in Section 7.

The headline breadth is sixteen landmark solution rows and thirteen quantitative outputs or checks from one shared source architecture. The deliberately generous four-cluster coincidence budget in Section 7 is 10^{-9}, about one in a billion; a four-decade search penalty gives 10^{-5}, about one in one hundred thousand.

Throughout, *derived* denotes an output of a declared source map whose inputs and branch rules are fixed independently of the comparison target. Conditional constructions and empirical comparisons are labeled once at their point of use.

16 landmark solutions, 13 quantitative outputs, 0 fitted decimals

The auditable zero-input statement is precise: after the five axioms and the declared discrete receipt branch are frozen, the counted rows introduce *zero target-fitted continuous parameters*. The closure coordinates are solutions,

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star),$$

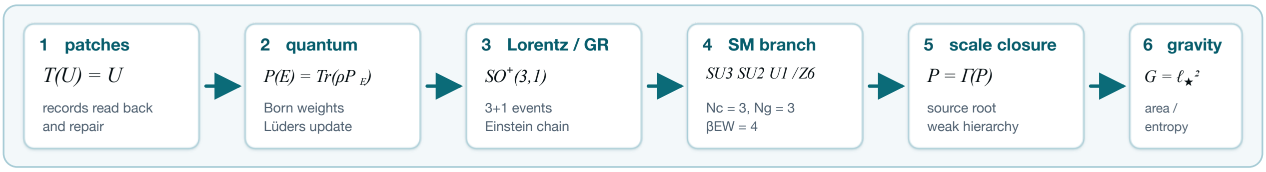
and the integers $(N_c, N_g, \beta_{\mathrm{EW}}) = (3, 3, 4)$ are branch outputs. Each sector reads the same source normal form without a separate adjustable decimal.

The positive proof spine

The compact argument follows six steps:

$$\begin{gathered}
\text{finite observer patches} \longrightarrow T(\mathfrak{U}) = \mathfrak{U} \longrightarrow \text{central record algebra}, \\
S^2 \text{ cap geometry} \longrightarrow \mathrm{SO}^+(3,1) \longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \\
\text{source closure} \longrightarrow (P_{\mathrm{src}}, N_\star), \\
\text{compact-gauge reconstruction} \longrightarrow \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \\
(P_{\mathrm{src}}, \alpha_U, 3, 3, 4) \longrightarrow \frac{v}{E_\star}, \qquad \frac{a_{\mathrm{cell}}}{4\bar{\ell}_{\mathrm{shared}}} \longrightarrow G_{\mathrm{geom}}.
\end{gathered}$$

The companion papers carry the full theorems and executable receipts.



1. The observer-patch fixed point

The mathematical object is a finite patch system. Patches expose ports, write records, compare overlaps, repair disagreement, and return a shared normal form [O1,O2]. Let T be the observer-overlap repair map and let $\mathfrak{U}_{\mathrm{OPH}}$ denote the selected normal form. The simulation identity is

$$T(\mathfrak{U}_{\mathrm{OPH}}) = \mathfrak{U}_{\mathrm{OPH}}.$$

The OPH simulation theorem establishes this identity from endogenous update, observer-readable records, schedule-independent overlap repair, selected-branch elimination, and implementation/clock closure [O2]. The universe is the stable record world returned by its own readback-and-repair computation.

Conditional closure theorem

Fix the OPH axioms, branch rules, and quotient/readout maps. Let

$$\Gamma_P : I_P \to I_P, \qquad \Gamma_N : I_N \to I_N$$

be source-only contractions selected uniquely by those rules up to observer-invisible equivalence. Banach's theorem gives unique endpoints

$$P_\star = \Gamma_P(P_\star), \qquad N_\star = \Gamma_N(N_\star).$$

Every declared observable

$$Y_i = \pi_i(\mathrm{nf}_{\mathrm{OPH}}(P_\star, N_\star))$$

is therefore fixed by the selected branch and invariant under admissible implementation and schedule changes. Data comparison identifies the physical branch represented by these source outputs.

The concrete carrier is a finite collection of local states, ports, central records, repair maps, and checkpoints. The reference benchmark replays all six overlap schedules and returns the same repaired normal form on the clean abelian branches [C5]. A separate Lean 4/Mathlib artifact certifies observer confluence, approximate schedule independence, refinement naturality, and the contractive conditional-resampling projector in 98 sorry-free theorems [C6].

For observer patch i, the readout equations are

$$r_i = \mathcal{R}_i(\mathfrak{U}_{\mathrm{OPH}}), \qquad \mathfrak{U}_{\mathrm{OPH}} = T(\{r_i\}_{\mathrm{compat}}).$$

The first map gives the local record; the second repairs compatible local records into the public normal form. This is the observer-like self-reading structure specific to OPH: local records participate in the computation that returns the world those records describe.

Quantum event surface

On a completed compare/write/verify slice, observer-accessible events generate a finite central record algebra. For an event E,

$$\mathbb{P}(E) = \mathrm{Tr}(\rho P_E), \qquad \rho|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

These are the Born rule and Lüders update. On the associated two-wing Bell surface,

$$|S_{\mathrm{CHSH}}| \le 2\sqrt{2}.$$

A finite 65,536-record run reproduces the event algebra numerically: committed-event weights sum to 1.0, record projectors have zero idempotency defect, and repeat reads are stable [O1,O3,C7].

2. Relativity, (3+1) events, and Einstein dynamics

The screen readout supplies an oriented conformal S^2. Its connected conformal group is the connected Lorentz group,

$$\mathrm{Conf}^+(S^2) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1).$$

The corresponding observer-frame space is

$$H^3 \cong \frac{\mathrm{SO}^+(3,1)}{\mathrm{SO}(3)}, \qquad \dim H^3 = 6 - 3 = 3.$$

On the declared event branch, repaired records populate a four-dimensional event base of signature $(-{+}{+}{+})$, while H^3 is the fiber of unit timelike observer frames over each event [O4].

The 2π-normalized modular flow of screen caps supplies the observer-relative clock. Half-sided modular inclusions give positive null translations; null tomography assembles their directional charges into a conserved local T_{ab}. Fixed-cap generalized-entropy stationarity, the uniform small-ball limit, and the all-directions tensor upgrade then compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\mathrm{geom}} \langle T_{ab} \rangle.$$

Every arrow in this chain has a named theorem and a machine-readable receipt type [O4,C8].

Latest simulation receipts

A transactional cap-net tower with 20/80/320 records reaches conflict-free, schedule-independent normal forms at every stage. On the associated Gaussian MaxEnt collar states, the 2π-BW profile residual improves from 4.4×10^{-3} to 1.1×10^{-4}. The realized bulk event packet has rank four and a seed-stable ancestry cone of signature $(1,3)$, with the modular clock aligned to its timelike direction at 0.999–1.000. These are finite/one-particle witnesses for every structural receipt family in the composed Einstein branch; the continuum result carries its declared scaling and physical-identification receipts [O4,C7,C8].

3. From closure coordinates to physical structure

The local pixel closure

The source-audit pixel candidate is

$$P_{\mathrm{src}} = 1.630972095694329\ldots.$$

A trial pixel P_k produces a field-width readback $R_P(P_k)$, which the pixel chart turns into the next iterate:

$$\alpha_k = R_P(P_k), \qquad P_{k+1} = \Gamma_P(P_k) = \varphi + \alpha_k\sqrt{\pi}, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

On the selected particle branch, the internal readback chain is

$$P \longmapsto \alpha_U(P) \longmapsto A_Z(P) \longmapsto A_T^{\mathrm{src}}(P), \qquad \alpha_{\mathrm{in}}(P) = \frac{1}{A_T^{\mathrm{src}}(P)}.$$

The intended closure equation would identify the geometric and internal readings:

$$P_{\rm src} = \varphi + \frac{\sqrt{\pi}}{A_T^{\rm src}(P_{\rm src})},$$
$$\alpha_{\rm root} = \frac{P_{\rm src} - \varphi}{\sqrt{\pi}} = \frac{1}{A_T^{\rm src}(P_{\rm src})}.$$

Bisection of $\rho(\alpha) = \alpha_{\rm in}(\varphi + \alpha\sqrt{\pi}) - \alpha$ records the candidate

$$\alpha_{\rm root}^{-1} = 136.994835164621649\ldots.$$

The two printed decimals have witness status. Inserting the inverse width gives $\varphi + \sqrt{\pi}/A_T^{\rm src} = 1.6309720958601054\ldots$, so the printed $P_{\rm src}$ has residual -1.65776×10^{-10}. The printed $P_{\rm src}$ implies inverse width $136.9948369199411\ldots$. The stored source-audit status is therefore a branch witness. A strict interval root and same-branch dependency certificate are work in progress. The source-audit solve is reproduced by

```
cd code/P_derivation && python3 derive_p.py --no-hadron-closure.
```

The measured low-energy endpoint is $\alpha^{-1}(0) = 137.035999177(21)$, leaving an offset of 0.0411640124 inverse-α units. Three empirical routes, the Thomson-endpoint interval, the standard on-shell decomposition, and the Ward-projected hadronic dispersion integral, locate that offset in the same low-energy transport sector with compatible magnitude and sign [C1,C2,S2].

The global capacity closure

Let $\Omega_N^{\rm sc}$ be the closed-screen normal forms supported at capacity N. The global source map compares their record capacity with the screen budget and applies Minimal Admissible Realization (MAR):

$$S(N) = \log|\Omega_N^{\rm sc}| - N, \qquad N_\star = \mathrm{MAR}\arg\max_N S(N).$$

The selected endpoint fixes the horizon normalization through

$$\Lambda_\star \ell_\star^2 = \frac{3\pi}{N_\star}, \qquad \Lambda_\star a_{\rm cell} = \frac{3\pi P_{\rm src}}{N_\star}, \qquad a_{\rm cell} = P_{\rm src}\ell_\star^2.$$

Thus the local loop records the candidate pair $(P_{\rm src}, \alpha_{\rm root})$, while the global loop gives the paired symbolic capacity and horizon readout $(N_\star, \Lambda_\star)$ [O1,O4].

The compact-gauge branch

The gauge derivation starts with individually transportable seeds selected by zero-obstruction transport. A common strict representative fixes the tensor category carried by those seeds. Surjective pullback functors and forgetful fibers construct its refinement limit, and Doplicher–Roberts/Tannaka reconstruction reads off the compact group. MAR selects the realized one-generation/one-Higgs matter package; anomaly cancellation, Yukawa invariance, and the central kernel then fix the hypercharge lattice and quotient [O4]. The result is

$$G_{\rm SM} = \frac{\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y}{\mathbb{Z}_6}, \qquad N_c = 3, \qquad N_g = 3.$$

The selected hypercharge lattice is

$$Q_i = (3,2)_{1/6}, \qquad u_i^c = (\bar{3},1)_{-2/3}, \qquad d_i^c = (\bar{3},1)_{1/3},$$
$$L_i = (1,2)_{-1/2}, \qquad e_i^c = (1,1)_1, \qquad H = (1,2)_{1/2}, \qquad i = 1,2,3.$$

The Higgs doublet is the Borel–Weil carrier

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2, \qquad \mathrm{Stab}(\langle H\rangle) = \mathrm{U}(1)_Q.$$

Its connected adjoint is

$$(8,1,0) \oplus (1,3,0) \oplus (1,1,0),$$

which fixes charge quantization and excludes mixed $(3,2,\pm 5/6)$ connected adjoint generators. The same branch supplies the electroweak integer

$$\beta_{\rm EW} = N_c + 1 = 4.$$

The QCD-free hierarchy witness

The source-audit hierarchy witness reuses the pixel candidate and the compact-gauge integers. Its matched unified-width coordinate is

$$\alpha_U(P_{\rm src}) = 0.04112424744557487.$$

The screen count begins with a triangulated S^2. For curvature charge $q_v = 6 - \deg(v)$, Euler's identity and $3F = 2E$ give

$$\sum_v q_v = 6V - 2E = 12.$$

Refinement-stable MaxEnt realizes the twelve units as twelve equivalent fivefold defects, and edge-center completion turns them into twelve central ports. Reversible write/check orientation doubles the gauge-algebra dimension:

$$m_{\rm rep} = 2\dim(\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)) = 2(8+3+1) = 24.$$

With local cell entropy $\ell_{\text{cell}} = P_{\text{src}}/4$ and $\beta_{\text{EW}} = 4$, the electroweak share of screen depth is

$$\Gamma_{\text{EW}} = \beta_{\text{EW}}\ell_{\text{cell}}\frac{\log(N/\pi)}{12} = \frac{P_{\text{src}}}{12}\log(N/\pi).$$

The corresponding transmutation law is

$$\frac{v}{E_\star} = P_{\text{src}}^{-1/2}\exp\left[-\frac{2\pi}{\beta_{\text{EW}}\alpha_U(P_{\text{src}})}\right] = 2.0198114150099223\times 10^{-17}.$$

This is the dimensionless source-audit witness. A GeV weak scale additionally requires an independently source-closed physical interpretation and normalization of $E_\star$. The exact parametric law and the 12/24 counts are theorem outputs; the numerical endpoint has the certificate class `source-audit branch witness` [O5,C3]. The weak scale is a settled-form readout with naturality defect

$$\epsilon_H = 0, \qquad \epsilon_H \in [0,0].$$

Electroweak comparison surface

The color-amplitude/loop comparison assumes

$$c = \frac{\sqrt{N_c}}{2} = \frac{\sqrt{3}}{2}, \qquad d = \frac{N_c}{2} = \frac{3}{2}.$$

With those two coefficients, the frozen forward map emits

$$(M_W, M_Z, m_H, m_t) = (80.3692168,\ 91.1880067,\ 125.1832854,\ 172.3219533)\,\text{GeV}.$$

Against the comparison values stored with the receipt, the four offsets are $(0.0013, 0.194, 0.484, 0.370)\sigma$. This conditional result identifies the charged channel with one color intertwiner and the neutral channel with the closed color loop [O5,C4]. It is a quadratic comparison model. Evidence class: conditional coordinate surface. The quotient-transport certificate, physical scale, running scheme, and complex-pole map are work in progress.

Charged-family carrier and conditional benchmark

The charged-family benchmark begins with the balanced Hermitian circulant carrier and the branch-count phase

$$\delta_e = \frac{\beta_{\text{EW}}}{2N_cN_g} = \frac{2}{9}, \qquad C_{\delta_e} = I + \frac{1}{\sqrt{2}}\left(e^{i\delta_e}R + e^{-i\delta_e}R^2\right), \qquad R^3 = I.$$

Diagonalization gives

$$r_k = 1 + \sqrt{2}\cos\left(\delta_e + \frac{2\pi k}{3}\right),$$

and, after ordering,

$$(r_e, r_\mu, r_\tau) = (0.0403499082192,\ 0.580211920148,\ 2.379438171633).$$

For $m_i = s_e r_i^2$, the trace identities $\operatorname{tr} C = 3$ and $\operatorname{tr} C^2 = 6$ yield

$$\frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{2}{3},$$

and the scale-free ratios are

$$\frac{m_\mu}{m_e} = 206.770315972729\ldots, \qquad \frac{m_\tau}{m_e} = 3477.47283710460\ldots.$$

Against the PDG-2026 pole-mass summary, the two ratios differ by approximately 9.83 ppm and 30.94 ppm [S8]. This row is a conditional retrodictive benchmark: its stated inputs are the balanced carrier and the phase above [O5,O6].

A separate exact geometric result begins with the screen-microphysics icosahedron. Its twenty outward-oriented faces form A_5/C_3, and each face stabilizer cycles three corners. Every A_5-equivariantly transported fiberwise Hermitian operator commuting with that cycle has the form

$$C = aI + bR + \bar{b}R^2, \qquad \lambda_k = a + 2|b|\cos\left(\arg b + \frac{2\pi k}{3}\right),$$

with a face-representative-independent unordered spectrum [O3,O5]. This exact carrier theorem supplies the geometric three-cycle used by charged-family source maps.

A target-informed continuation on this carrier gives

$$(m_e, m_\mu, m_\tau) = (0.510998950843394,\ 105.658375501481,\ 1776.930000014351)\,\text{MeV}$$

on one hybrid tuple. Its 0.000300-ppm headline uses frozen rounded fields; the tau residual is -1.387289 ppm against the packet's higher-precision central value. A conditional theorem proves that the declared three-coordinate affine repair map is a strict contraction with one fixed point. Source-multiplier witnesses keep the same displayed Jacobian and contraction across different mass tuples. Evidence class: exact carrier plus a conditional declared-map theorem. Source selection and the physical pole map are work in progress [O5,O6].

4. Geometric gravity normalization and the SI readout

On the declared Einstein branch, the edge-entropy/area-law theorem identifies the coefficient of the local Einstein equation

$$G_{ab} + \Lambda g_{ab} = 8\pi G_{\text{geom}}\langle T_{ab}\rangle$$

and its Newton–Poisson limit with the observer-cell ratio

$$G_{\text{geom}} = \frac{a_{\text{cell}}}{4\bar{\ell}_{\text{shared}}}.$$

The local pixel branch supplies

$$a_{\text{cell}} = P_{\text{src}}\ell_\star^2, \qquad \bar{\ell}_{\text{shared}} = \frac{P_{\text{src}}}{4}.$$

Substitution gives

$$\frac{G_{\text{geom}}}{\ell_\star^2} = \frac{P_{\text{src}}}{4(P_{\text{src}}/4)} = 1, \qquad \boxed{G_{\text{geom}} = \ell_\star^2}.$$

The pixel number cancels because it counts both cell area and shared edge entropy. The factor 4 follows from the ratio between the structural Einstein coefficient and the Newton–Poisson convention [O4]. This result is the OPH normalization in geometric units.

Newton coupling and the SI clock readout

A cesium clock translates the geometric identity into SI units:

$$\gamma_\star = \frac{\ell_\star \nu_{\text{Cs}}}{c} = \frac{\varepsilon_{\text{Cs}}}{2\pi}, \qquad G_{\text{SI}} = \frac{c^3\ell_\star^2}{\hbar} = \frac{c^5\varepsilon_{\text{Cs}}^2}{4\pi^2\hbar\nu_{\text{Cs}}^2}.$$

For the selected display coordinate $\gamma_\star = 4.955974336548419\ldots \times 10^{-34}$, this gives

$$\boxed{G_{\text{SI}} = 6.674299995910528 \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}}.$$

The 2022 CODATA value is $6.67430(15) \times 10^{-11}\,\text{m}^3\,\text{kg}^{-1}\,\text{s}^{-2}$, so the display lies $2.7 \times 10^{-5}\sigma$ from its central value [S1]. Evidence class: exact geometric normalization plus conditional SI scale/clock display. The source-promotion certificate is the cesium record stack $\mathcal{R}_U + \mathcal{R}_\alpha + \mathcal{R}_e^{\text{abs}} + \mathcal{R}_{\text{QCD/nuc}}^{133\text{Cs}} + \mathcal{R}_{\text{atom}}^{133\text{Cs}}$ [O1,O4,C3].

5. Landmark solution map

The table states the positive result first and records its evidence class once. Paper numbers refer to the numbered OPH papers in the References block.

Physics problem	OPH solution on the declared branch	Evidence class	Paper
Observer and measurement problem	An observer is a bounded self-reading record boundary. Compatible local readouts participate in the repair map that returns the public world $T(\mathfrak{U}) = \mathfrak{U}$.	Internal theorem	1,2
Quantum probabilities and update	The finite central record algebra yields the Born rule, Lüders conditioning, and $\lvert S_{\text{CHSH}}\rvert \le 2\sqrt{2}$ on the two-wing event surface.	Internal theorem	1,3
Time, Lorentz symmetry, and three space dimensions	Cap modular flow supplies the relative clock; $\text{Conf}^+(S^2) \cong \text{SO}^+(3,1)$, and $H^3 \cong \text{SO}^+(3,1)/\text{SO}(3)$ has dimension three over a $3+1$ event base.	Receipt-conditional theorem	4
General relativity	Null translations, local stress, generalized-entropy stationarity, and the uniform small-ball limit compose to $G_{ab} + \Lambda g_{ab} = 8\pi G\langle T_{ab}\rangle$.	Receipt-conditional theorem	4
Gravity from quantum information	The edge-entropy/area law fixes the Einstein and Newton coefficient: $G_{\text{geom}} = a_{\text{cell}}/(4\bar{\ell}_{\text{shared}}) = \ell_\star^2$.	Internal identity	4
Standard Model gauge and matter structure	DR/Tannaka reconstruction plus MAR, anomaly cancellation, and Yukawa invariance give $(\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$, the hypercharge lattice, $N_c = N_g = 3$, and one Higgs doublet.	Realized-branch theorem	4
Electroweak breaking and charge quantization	The Borel–Weil Higgs carrier $H^0(\mathbb{CP}^1, \mathcal{O}(1)) \cong \mathbb{C}^2$ has stabilizer $\text{U}(1)_Q$; the connected adjoint fixes the Standard Model charge lattice.	Realized-branch theorem	4
Electromagnetism and force carriers	The unbroken $\text{U}(1)_Q$ branch fixes the electromagnetic connection. With the declared quadratic actions, Maxwell has two transverse modes, perturbative Yang–Mills has $2\dim G$, and Einstein gravity has two transverse-traceless modes.	Conditional action theorem	1,4
Proton stability	The selected product-group gauge structure contains no grand-unified gauge bosons connecting quarks to leptons and therefore excludes gauge-mediated proton decay on the realized branch.	Realized-branch corollary	1,4
Weak/UV hierarchy and Higgs naturality	The 12-port screen, 24-slot oriented register, P_{src}, and $\alpha_U(P_{\text{src}})$ emit $v/E_\star \sim 2.02 \times 10^{-17}$ with $\epsilon_H = 0$.	Source-audit witness	5
Fine-structure source scale	The local pixel closure emits P_{src} and $\alpha_{\text{root}}^{-1} = 136.9948351646\ldots$; independent hadronic transport has the location, magnitude, and sign of the low-energy offset.	Source-audit witness	5,6
Dark energy / cosmological term	The global record-capacity closure fixes $\Lambda_\star\ell_\star^2 = 3\pi/N_\star$, while vacuum energy lies in the kernel of the local null readout.	Symbolic closure theorem	1,4
Dark-matter phenomenology	Repair-stress bookkeeping gives the galaxy continuation $\nu_{\text{OPH}}(x) = [1 - e^{-\lambda_{\text{collar}}\sqrt{x}}]^{-1}$ and the deep limit $v^4 = GM_b a_{\text{eff}}$.	Conditional continuation	7
Yang–Mills gap	On the controlled compact-gauge continuum branch, the transfer/repair theorem identifies $\Delta_{\text{YM}} = \Delta_{\text{rep}} > 0$.	Conditional theorem	8
Family geometry	The icosahedral face bundle A_5/C_3 gives an exact local cyclic carrier; the balanced $\delta_e = 2/9$ continuation yields the charged-lepton scale-free ratios.	Exact carrier + conditional map	3,5,6
String-vacuum selection	Observer-patch consistency defines a finite sieve over critical-string candidates; the Bouchard–Donagi geometry is selected on its declared gate stack.	Conditional selection program	9

6. Best quantitative outputs and checks

Output	Evidence class	Value and check	Paper / receipt
Finite observer-patch normal form	Internal theorem	All six overlap schedules return the same repaired normal form on the clean abelian branches, with zero schedule divergence.	1,2 / C5
Quantum event surface	Internal theorem	On 65,536 records, Born weights sum to 1.0, projector idempotency defect is zero, and repeat-read fraction is 1.0; analytically $\lvert S_{\text{CHSH}}\rvert \le 2\sqrt{2}$.	1,3 / C7

Output	Evidence class	Value and check	Paper / receipt
Lorentz and Einstein chain	Receipt-conditional theorem	BW residual improves $4.4 \times 10^{-3} \to 1.1 \times 10^{-4}$; the realized bulk packet has rank four, signature $(1, 3)$, and clock alignment 0.999–1.000.	4 / C8
Standard Model branch	Receipt-conditional realized theorem	Exact quotient $\mathbb{Z}_6$, one Higgs doublet, and $(N_c, N_g, \beta_{\text{EW}}) = (3, 3, 4)$, matching the observed discrete architecture.	4
Classical force content	Conditional action theorem	Two transverse Maxwell modes, $2 \dim G$ perturbative pure-Yang–Mills modes, and two transverse-traceless Einstein modes, all with the common massless quadratic kernel on their declared action branches.	1,4
Source electromagnetic root	Source-audit numerical witness	$P_{\text{src}} = 1.630972095694329\ldots$ and $\alpha_{\text{root}}^{-1} = 136.994835164621649\ldots$; the source-to-CODATA relative offset is 3.00×10^{-4}.	5,6 / C1
EW-refined capacity	Public-endpoint bridge fixed point	$N_{\text{CRC}}^{\text{EW}} = 3.5323546226929907 \times 10^{122}$; its logarithmic depth differs from the rounded 3.31×10^{122} de Sitter scale by 2.1×10^{-4}.	5 / C3
Weak hierarchy	Source-audit numerical witness	$\alpha_U(P_{\text{src}}) = 0.04112424744557487$ and $v/E_\star = 2.0198114150099223 \times 10^{-17}$, with $\epsilon_H = 0$.	5 / C3
Electroweak mass surface	Conditional comparison surface	The color-amplitude/loop ansatz gives the comparison coordinates $(M_W, M_Z, m_H, m_t) =$ $(80.3692, 91.1880, 125.1833, 172.3220)$ GeV; evidence class: conditional coordinate comparison.	5 / C4
Charged-family geometry	Exact carrier + conditional declared-map theorem	The A_5/C_3 face bundle gives local cyclic fibers; the balanced $\delta_e = 2/9$ benchmark gives $m_\mu/m_e = 206.77031597\ldots$ and $m_\tau/m_e = 3477.47283710\ldots$, differing by 9.83 and 30.94 ppm.	3,5,6
Geometric gravity	Internal identity	The cell-area/shared-entropy theorem gives the exact normalization $G_{\text{geom}}/\ell_\star^2 = 1$.	4
SI gravity readout	Conditional scale/clock display	$G_{\text{SI}} = 6.674299995910528 \times 10^{-11}$ in SI units, lying $2.7 \times 10^{-5}\sigma$ from the CODATA central value.	1,4 / S1
Galaxy scorecard	Conditional galaxy continuation	SPARC χ^2/point = 1.017, velocity RMS 11.25 $\text{km}\,\text{s}^{-1}$, and RAR scatter 0.13 dex; same-function reference χ^2/point = 1.023.	7 / C9

7. Why coincidence is unlikely

One source, every major layer

One finite self-reading architecture reaches quantum theory, $3+1$ spacetime, Einstein gravity, the Standard Model, the weak hierarchy, cosmic closure, and Newton normalization. Sector-specific fitted decimals in this core: 0.

Core event	Generous window	Factor
The same patch rules support Lorentz $3+1$ kinematics and the Einstein chain	one chance in ten	10^{-1}
The realized compact branch lands on the exact Standard Model quotient, hypercharge/matter lattice, three colors, and three generations	one chance in one thousand	10^{-3}
The target-separated electromagnetic source root lands within 3.0×10^{-4} relative of the low-energy endpoint, with the offset in the independently known transport sector	one chance in one thousand	10^{-3}
The source-audit hierarchy exponent lands in the observed 10^{-17} weak sector with $\epsilon_H = 0$	one chance in one hundred	10^{-2}

$$p_{\text{coincidence}} = (10^{-1})(10^{-3})(10^{-3})(10^{-2}) = 10^{-9}, \qquad -\log_2 p_{\text{coincidence}} = 29.9 \text{ bits.}$$

Give OPH and coincidence equal odds before reading the table, use the product as the likelihood of the evidence under coincidence, and apply a 10^4 search penalty. The simple Bayes score is

$$\boxed{\Pr(\text{OPH incorrect} \mid E) \approx \frac{10^4 p_{\text{coincidence}}}{1 + 10^4 p_{\text{coincidence}}} = 0.001\%},$$

about one chance in 100,000. A calibrated posterior would require measured likelihoods and a defensible prior; this score keeps its assumptions visible.

References

Data and standards

[S1] CODATA Task Group (2024). "2022 CODATA recommended values: Newtonian constant of gravitation." NIST Reference on Constants.

[S2] CODATA Task Group (2024). "2022 CODATA recommended values: inverse fine-structure constant." NIST Reference on Constants.

[S8] Particle Data Group (2026). "Leptons." Review of Particle Physics summary table.

OPH papers All OPH series papers below use release `r1534`.

[1/O1] Mueller et al. (2026). "Observers Are All You Need."

[2/O2] Müller et al. (2026). "Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics."

[3/O3] Müller et al. (2026). "Federated Echosahedral Screen Microphysics: Patch Hardware, Records, and Observer Synchronization in OPH."

[4/O4] Müller et al. (2026). "Recovering Relativity and the Standard Model from Observer Overlap Consistency."

[5/O5] Müller et al. (2026). "Deriving the Particle Zoo from Observer Consistency."

[6/O6] Müller, B. (2026). "The Fine-Structure Constant as an OPH Pixel Fixed Point." OPH technical note.

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[8/O8] Müller, B. (2026). "Explaining the Yang–Mills Mass Gap with Observer-Patch Repair Dynamics." OPH technical note.

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Executable artifacts

[C1] FloatingPragma (2026). Pixel-closure solver and receipts.

[C2] FloatingPragma (2026). Particle-branch source and audit suite.

[C3] FloatingPragma (2026). Hierarchy theorem package and receipts.

[C4] FloatingPragma (2026). Electroweak calibration receipts.

[C5] FloatingPragma (2026). Consensus schedule benchmarks.

[C6] FloatingPragma (2026). Lean 4/Mathlib theorem artifact.

[C7] Mueller, B. (2026). OPH-FPE simulation and receipts, including the quantum-record, BW/KMS, H^3, and geometry runs.

[C8] FloatingPragma (2026). Lorentz–Einstein geometry receipts.

[C9] FloatingPragma (2026). SPARC galaxy scorecard and receipts.